HRMS & Payroll Management System - Complete Knowledge Transfer (KT) Report

Document Title: System Operations, Shift Assignment, Employee Onboarding & Database Taxonomy
Target Audience: Development Team, HR Administrators, System Administrators & Technical Lead
System Name: HRMS & Payroll Central Platform

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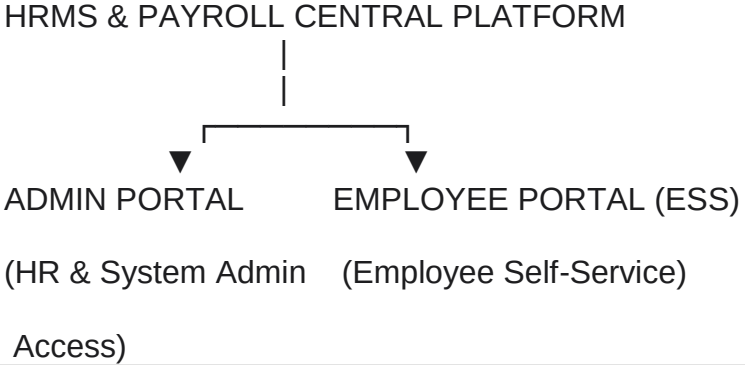
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1. Overview & System Architecture

The HRMS platform operates as a dual-portal system:

- **Admin Portal:** System & HR configurations, employee onboarding, shift management, leave approvals, payroll processing, and statutory tax returns.

Employee Self-Service (ESS) Portal: Daily web/mobile attendance clocking, leave requests, reimbursement claims, payslip downloads, and tax proof submissions.



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2. Troubleshooting Attendance Swiping

Error Message: `"Failed to update attendance status"`
When an employee logs into the ESS Portal and clicks **CLOCK IN / CLOCK OUT**, the frontend invokes `/api/v1/attendance/check-in` or `/api/v1/attendance/check-out`.

Root Causes & Solutions:

Root Cause	Technical Reason	Solution / Fix
Email Mismatch between User & Employee	`AttendanceService.GetEmployeeForUserAsync` links `User.Email` to `Employee.Email`. If the User account email is `john@company.com` and Employee email is `john.doe@company.com`, the lookup returns `null`.	Ensure the login `User.Email` matches the `Employee.Email` field **exactly** in the database / Admin Portal.
Missing Employee Profile	User account was created in `users` table without a corresponding record in the `employees` table.	Onboard the employee in Admin Portal under **Employees -> Add Employee** before creating the user account.
Inactive Employee Status	The employee's `status` column is set to `"Inactive"` or `is_deleted = true`.	Change employee status to `"Active"` in Employee Profile settings.
Missing Shift Assignment	No active shift assignment (`shift_assignments`) is configured for the employee for the current date.	Assign a Shift to the employee (see Section 3).
Geofencing / Location Access Block	`AttendancePolicy` has `EnableGeofencing = true`, but browser location permissions are denied or coordinates are outside the geofence radius.	Grant browser location access or disable mandatory geofencing in `AttendancePolicy`.

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3. Shift Assignment Procedure

Shifts define working hours, grace periods, half-day thresholds, and week-offs.

Step 1: Create a Shift Master (Admin Portal / API)

1. Go to **Admin Portal -> Settings / Attendance -> Work Shifts**.
2. Click **Create New Shift**.
3. Fill in shift details:
   - **Shift Name:** `General Shift`
   - **Start Time:** `09:00 AM` | **End Time:** `06:00 PM`
   - **Grace Period:** `15 minutes` (Late after 9:15 AM)
   - **Half-Day Threshold:** `4.5 hours`
   - **Is Active:** `True`
4. Click **Save Shift**.

Step 2: Assign Shift to an Employee

1. Go to **Admin Portal -> Attendance -> Shift Assignments** (or **Employee Management -> View Employee -> Shift**).
2. Click **Assign Shift**.
3. Select:
   - **Employee:** Select the newly created employee.
   - **Shift:** Select `General Shift`.
   - **Effective From:** Set to date of joining or current date.
   - **Effective To:** Leave blank for permanent assignment.
   - **Is Active:** `True`.
4. Click **Save Assignment**.

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4. New Employee Mandatory Settings Checklist

To ensure a new employee can log in, view their dashboard, and successfully mark attendance, verify all 6 mandatory configurations:

[1] Employee Profile Created (Employees Table)
  |
[2] User Login Account Linked (Users Table - Email Must Match Exactly)
  |
[3] Account Role Assigned (UserRoles Table -> Role: 'Employee')
  |
[4] Employee Status Active (Status = 'Active')
  |
[5] Active Shift Assigned (ShiftAssignments Table -> Effective Today)
  |
[6] Attendance Policy Configured (AttendancePolicies Table -> Default Active Policy)

1. **Exact Email Linkage:** `User.Email` **MUST EQUAL** `Employee.Email` (Case-insensitive).
2. **Active Employee Status:** `employees.status = 'Active'` and `employees.is_deleted = false`.
3. **Assigned Shift:** At least one active row in `shift_assignments` where `effective_from <= today` and (`effective_to >= today` or `NULL`).
4. **Assigned Organization Access:** `user_organization_accesses` row linking `UserId` to `OrganizationId`.
5. **Role Assignment:** `user_roles` linking `UserId` to the `Employee` role.
6. **Leave Balances Provisioned:** Initialized records in `leave_balances` for Casual Leave, Sick Leave, and Earned Leave.

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5. Database Clean Setup & Table Taxonomy

When preparing a **fresh/clean DB environment**, it is critical to distinguish between **Master Configuration Tables** (which hold application rules and must **NEVER** be deleted) and **Operational / Transactional Tables** (which store employee transactions and can be wiped).

Category A: Basic Configuration & Master Tables (DO NOT DELETE ●)

These tables store application metadata, system rules, tax slabs, menu structures, and calculation formulas.

Table Name	Description / Purpose
organizations	Corporate entity profile and tax registration codes (PF, ESI, TAN, PT)
branches	Office locations and GST registrations
departments	Company departments (Engineering, HR, Finance, etc.)
roles	System security roles (Admin, HR, Finance, Employee, Manager)
permissions	Granular permission action keys
role_permissions	Security mappings between roles and permissions
menu_items	Navigation menu structure for Admin & Employee portals
shifts	Work shift definitions (timings, grace periods, week-offs)
attendance_policies	Attendance rules (late penalties, geofence settings, web punch options)
overtime_policies	Overtime rate multipliers and approval policies
salary_components	Component master (Basic, HRA, PF, ESI, PT, Allowances)
salary_structures	Salary templates (Grade A, Grade B CTC formulas)
salary_structure_components	Link between templates and salary components
pay_schedules	Monthly pay run cycles and cutoff dates
leave_types	Leave categories (Casual Leave, Sick Leave, Earned Leave)
leave_settings	Carry-forward, encashment, and accrual policies
holidays	Company annual public holiday calendar
tax_year_configurations	Financial year income tax slab rules
tax_year_settings	Tax regime default choices and limits
professional_tax_configurations	State-wise Professional Tax (PT) slab rules
statutory_deduction_configurations	EPF & ESI ceiling limits and wage rates
organization_tax_configurations	Per-org tax settings
compliance_rule_versions	Statutory compliance versioning (Labour Codes 2025)
lwf_configurations	State-wise Labour Welfare Fund rules
calculation_rule_sets	Dynamic payroll calculation engine rules
calculation_rule_versions	Rule set version history

Category B: Operational & Transactional Data Tables (CAN BE CLEARED FOR CLEAN DB □)

These tables contain employee records, login sessions, daily logs, and monthly pay runs.

Table Name	Description / Content
users	User login credentials (Keep only 1 *SuperAdmin* account!)
user_roles	User to role mappings
user_organization_accesses	User to organization access scopes
refresh_tokens	Active JWT refresh token sessions
user_login_attempts	Audit log of login attempts
employees	Employee personal, bank, CTC, and profile details
employee_salary_structures	Employee assigned CTC salary breakdowns
shift_assignments	Employee shift roster assignments
attendance_logs	Daily employee check-in / check-out time punches
attendance_regularizations	Missed punch correction requests
overtime_requests	Employee overtime claim submissions
leave_requests	Applied leave transactions
leave_balances	Calculated leave quotas for employees
leave_encashments	Leave encashment claims
employee_tax_declarations	Employee 80C/80D tax investment declarations
payroll_runs	Monthly payroll execution header records
employee_payroll_runs	Monthly itemized salary slips
payrun_validation_snapshots	Pre-payroll validation audit checks
payrun_payment_configurations	Pay run disbursement modes
payrun_payment_validation_snapshots	Bank payment pre-checks
payrun_attendance_snapshots	Snapshot of attendance during pay run
payrun_attendance_snapshot_items	Itemized attendance days per employee
payrun_locks	Payroll period lock status
payslip_send_logs	Log of emailed payslips
payrun_summaries	Summary totals for financial reports
payrun_audit_logs	Audit trail of payroll actions
payrun_rollbacks	Rollback audit history
off_cycle_payroll_runs	Off-cycle / bonus pay runs
off_cycle_payroll_run_employees	Employees in off-cycle runs
final_settlements	Resigned employee Full & Final settlements
variable_pay_entries	Monthly bonus / commission entries
reimbursement_claims	Expense claims submitted by employees

Table Name	Description / Content
reimbursement_payments	Payout history for expenses
employee_loans	Active salary loans & advances
employee_loan_repayments	Monthly loan EMI deduction history
announcements	Company announcements broadcasted to employees